



Geochimica et Cosmochimica Acta

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Before you begin

Geochimica et Cosmochimica Acta (GCA) publishes research papers in a wide range of subjects in **terrestrial geochemistry**, **meteoritics**, and **planetary geochemistry**. GCA aims to present studies of fundamental significance

and broad relevance for understanding geochemical systems, mechanisms, and processes, and of interest to a broad and diverse audience of geochemists. The scope of the journal includes:

1. Physical chemistry of gases, aqueous solutions, glasses, and crystalline solids
2. Igneous and metamorphic petrology
3. Chemical processes in the atmosphere, hydrosphere, biosphere, and lithosphere of the Earth
4. Organic geochemistry
5. Isotope geochemistry
6. Meteoritics and meteorite impacts
7. Lunar science; and
8. Planetary geochemistry.

If you would like more information about the Geochemical Society and Meteoritical Society, visit their Home Pages at <http://www.geochemsoc.org> and <http://www.meteoriticalsociety.org>

Types of papers

Geochimica et Cosmochimica Acta publishes original research articles, comments and replies on papers previously published in the journal. Citations for awards given by the sponsoring societies are also published in GCA. The journal does not publish papers focused on method, technique, or model development unless accompanied by a novel scientific result enabled by the new development that has demonstrated broader geochemical significance. In addition, GCA does not publish regional case studies. Investigations of field sites that use the results to provide new insight into fundamental geochemical processes are certainly welcome and encouraged.

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Comment and Reply submissions to GCA should be short (not exceeding 2000 words), concise discussions of major, substantive aspects of the original article. Comments will only be entertained for publication if they significantly impact the interpretation of the original article by exposing serious flaws in reasoning, deficiencies in experimental design, or other factors that significantly impact the original article's wider usage by the GCA readership. Discussion of minor errors or omissions, or presentation of alternative points of view, are not suitable for Comments. New or unpublished data or results are not to be presented as either Comments or Replies but should be submitted as research articles. The paper to which the Comment is directed should have been published in GCA within the previous 12 months. Comments and Replies should contain no more than 1 figure and 10 references; excessive use of multi-part figures is not permitted and the Executive Editor will make a decision on the suitability of such figures. Authors who are considering the submission of a Comment to GCA are encouraged to contact the editorial office (gca@elsevier.com) for additional information.

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Please see our information on [Ethics in publishing](#).

Declaration of competing interest

All authors must disclose any financial and personal relationships with other people or organizations that could inappropriately influence (bias) their work. Examples of potential conflicts of interest include employment, consultancies, stock ownership, honoraria, paid expert testimony, patent applications/registrations, and grants or other funding. Authors should complete the declaration of competing interest statement using [this template](#) and upload to the submission system at the Attach/Upload Files step. **Note: Please do not convert the .docx template to another file type. Author signatures are not required.** If there are no interests to declare, please choose the first option in the template. [More information](#)

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Reporting guidance

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Definitions

Sex generally refers to a set of biological attributes that are associated with physical and physiological features (e.g., chromosomal genotype, hormonal levels, internal and external anatomy). A binary sex categorization (male/female) is usually designated at birth ("sex assigned at birth"), most often based solely on the visible external anatomy of a newborn. Gender generally refers to socially constructed roles, behaviors, and identities of women, men and gender-diverse people that occur in a historical and cultural context and may vary across societies and over time. Gender influences how people view themselves and each other, how they behave and interact and how power is distributed in society. Sex and gender are often incorrectly portrayed as binary (female/male or woman/man) and unchanging whereas these constructs actually exist along a spectrum and include additional sex categorizations and gender identities such as people who are intersex/have differences of sex development (DSD) or identify as non-binary. Moreover, the terms "sex" and "gender" can be ambiguous--thus it is important for authors to define the manner in which they are used. In addition to this definition guidance and the SAGER guidelines, the [resources on this page](#) offer further insight around sex and gender in research studies.

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Cover Letter

All manuscripts must be accompanied by a Cover Letter. This document should briefly explain how the manuscript addresses the specific aim of the journal: to present studies of fundamental significance and broad relevance for understanding geochemical systems, mechanisms, and processes, and of interest to a broad and diverse audience of geochemists. The letter should also address any potential conflicts of interest that the editors should be aware of as they consider the submission. Authors may include requests that specific individuals be avoided as editors or reviewers, although at GCA such requests are considered purely advisory in nature and are not binding. Finally, authors are also required to provide in the Cover Letter the **names of three Associate Editors** who may have the requisite expertise to manage the assessment of the manuscript. Note that lack of availability and conflicts of interest may require that a different Associate Editor handle the manuscript. The list of currently active Associate Editors can be found at: <https://www.sciencedirect.com/journal/geochimica-et-cosmochimica-acta/about/editorial-board>.

Suggested Reviewers

During the submission process authors are requested to provide the name, affiliation, and contact information for five potential reviewers. These scientists should be knowledgeable in key aspects of the manuscript topic. Suggestions for scientists at all career stages and from diverse backgrounds are welcome and encouraged. GCA seeks to have reviewers that reflect the diversity, broadly defined, of our authors and readers.

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GCA considers editors and reviewers to have a conflict of interest with a manuscript if they have collaborated with any of the authors within the last three years. There are cases where such associations do not pose a conflict, such as when two scientists are part of a large, multi-institutional team but otherwise do not collaborate extensively. Such exceptions are evaluated by the editors. Colleagues at the same institution and recent advisors or advisees are also considered to have a conflict of interest. Other scientists working on the same research topic are not considered to be conflicted unless they have previously displayed bias.

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Preparation

Manuscript Preparation and Layout

Use of word processing software

It is important that the file be saved in the native format of the word processor used. The text should be in single-column format. Keep the layout of the text as simple as possible. The electronic text should be prepared in a way very similar to that of conventional manuscripts (see also the [Guide to Publishing with Elsevier](#)).

To avoid unnecessary errors you are strongly advised to use the 'spell-check' and 'grammar-check' functions of your word processor.

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Line Numbers

Line numbers should be added to your file, numbered continuously throughout the main text.

Sections

Divide your article into clearly defined and numbered sections. Subsections should be numbered 1.1 (then 1.1.1, 1.1.2, ...), 1.2, etc. (the abstract is not included in section numbering). Use this numbering also for internal cross-

referencing: do not just refer to 'the text'. Any subsection may be given a brief heading. Each heading should appear on its own separate line.

Appendices

Appendices are no longer used by GCA. Content intended for an appendix should instead be placed in the Supplementary Material. Authors should not include an Appendix section in their manuscript.

Essential title page information

Title.

The title should be concise, informative, and helpful for identifying the key purpose of the article in during electronic searches. Title should convey the major scientific finding and not the methods used. Statement of the location of study is strongly discouraged as the journal does not publish regional case studies that do not address the primary aim of GCA. Avoid abbreviations and formulae where possible. GCA does not accept titles written in the form of a question.

Author names and affiliations.

Please clearly indicate the given name(s) and family name(s) of each author and check that all names are accurately spelled. You can add your name between parentheses in your own script behind the English transliteration. Present the authors' affiliation addresses (where the actual work was done) below the names. Indicate all affiliations with a lower-case superscript letter immediately after the author's name and in front of the appropriate address. Provide the full postal address of each affiliation, including the country name and, if available, the e-mail address of each author. Note that GCA does not allow the inclusion of deceased authors. All authors must confirm their agreement to be listed on the paper, including after revision, and this is not possible if an author is deceased.

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Clearly indicate the single person who will handle correspondence at all stages of refereeing and publication, also post-publication. This responsibility includes answering any future queries about the contents of the article by members of the scientific community. **Ensure that the e-mail address is given and that contact details are kept up to date by the corresponding author. GCA only allows one corresponding author to be identified**

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Manuscript Components

General Organization of Manuscripts

Manuscripts submitted to GCA are recommended to follow the following structure:

Title, Authors, and Affiliations

Abstract

Keywords

Introduction

Materials and Methods

Results

Discussion

Conclusions

Acknowledgments

Data Availability

Appendix A. Supplementary Material (*if the submission contains supplementary material*)

References

Tables

Figure Captions

Figures

Authors are free to embed figures and table throughout the manuscript if they choose. To avoid duplication, it is important that Authors do not upload copies as separate files for any embedded figures or tables. **Figures and Tables must only occur in one location.** Please use double-spaced 12 pt serif font (e.g., Times New Roman) or 11 pt sans serif font (e.g., Arial, Helvetica) for all main text sections of the manuscript. This ensures readability and ease of note taking should an editor or reviewer prefer to work with a printed form of the submission.

Abstract

A concise and factual abstract is required. The abstract should be one to two paragraphs in length limited to 500 words; 200-300 words is preferred. The abstract should briefly state the purpose of the research, the principal results, and major conclusions. An abstract is often presented separately from the article, so it must be able to stand alone. For this reason, references should be avoided, but if essential, then cite the author(s) and year(s). Also, non-standard or uncommon abbreviations should be avoided, but if essential they must be defined at their first mention in the abstract itself.

Keywords

Provide three to five keywords that succinctly identify the topic of the article or the methods used.

Introduction

The introduction should be a single section, without subsections, and should identify the primary motivation for the study. It should review the literature to justify the scientific question being addressed and the hypothesis to be tested, but this review should not be excessive and must focus only on topics of direct relevance to the study. It should conclude with a single paragraph outlining how the hypothesis or scientific question was investigated but should not summarize the results.

Materials and Methods

This section should provide enough information so that any reader with relevant expertise can reproduce the study. This includes the location coordinates of any field samples analyzed and all procedures involved in conducting experiments, making measurements, and data analysis. When appropriate, information on standards, quality control, and error propagation should be reported. Details for specific analyses should follow the conventions established by the field.

Results and Discussion

Results should be clearly distinguished from Discussion, and should be written as separately numbered sections in the manuscript. The Results should provide a concise description of the data or outcomes of the measurements and experiments presented in the manuscript, with adequate reference to tables and figures. Information provided as Results should form the observational basis for the Discussion and Conclusions. The Discussion should summarize and explain the main findings in an integrated and coherent manner, place the new results in the context of prior work to identify additional insights gained from the study, and address the broader geochemical significance and implications of the study.

In limited cases, combination of the Results section and Discussion section into a single "Results and Discussion" may be necessary to clearly convey the contents of the manuscript. Authors may petition the editors to allow for a combined "Results and Discussion" section in their Cover Letter at the time of submission. Combining these components into a single section requires specific justification beyond simply the writing style preference of the authors. **Note that all requirements for Discussion sections, including addressing the broader geochemical significance and implications of the study, are still required when Results and Discussion are combined.**

Conclusions

Inclusion of a Conclusions section is encouraged. It should not introduce substantial new ideas or integrate the results of the study with prior work from the literature, which should instead be presented in a Discussion section. If the conclusions simply summarize the key findings, then the section heading "Summary" is an acceptable alternative to the Conclusions.

Acknowledgements

Collate acknowledgements in a separate section at the end of the article before the references and do not, therefore, include them on the title page, as a footnote to the title or otherwise. List here those individuals who provided help during the research or writing of the article. In addition, authors should provide information on the sources of funding that supported the research, including the funding agency or foundation and, if available, the grant or award number. Funding provided to only one or a subset of authors may be indicated as such using the initials of the author(s), e.g., "(to XYZ)".

If no funding has been provided for the research, please include the following sentence: This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

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Authors are encouraged to include additional information that supports their research findings in an online Supplementary Material section. While this many consist of more than one file, Supplementary Material should be organized and collated into the smallest number of files possible. Supplementary files supplied will be published online alongside the electronic version of your article. Note that these files will be published in the exact format provided during submission, although the original filenames will not be preserved. The supplementary files should not contain tracked changes or other information that authors do not wish to have released publicly. If authors prepare one or more Supplementary Material files, then they should include in their manuscript a section with the heading "Appendix A. Supplementary Material". A brief summary of the contents of the file(s) included in Supplementary Material should be included beneath this heading. This section should be inserted immediately before the References. **Note that this small section is the only allowed usage of the term "Appendix". The Supplementary Material is not an appendix and cannot be referred to as such.**

Text, figures, and tables that support the article but are not necessary to convey the key findings of the work should be placed in a single PDF document. This should have a cover page that starts with the heading "Supplementary Material" and includes the article title and author list. Figures and tables should be numbers, with each preceded by an "S", e.g., Figure S1, Table S2, etc. These should be sorted and numbered in order of citation. Figures and tables should include appropriate captions, and additional references may be inserted into the supporting file using the same format as for the main article. Ideally, authors should prepare only one such supplementary document; there are no page limits. Please note that **GCA does not accept supplementary documents in DOCX or DOC format** because these will not directly render in web browsers. PDF format is required for all supplementary documents.

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GCA previously used the term "Electron Annex" for the types of supporting information files described above. Use of this term has ended and all such files should now be referred to as Supplementary Material.

Policies on Manuscript Components

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New meteorite names must be approved by the Meteorite Nomenclature Committee of the Meteoritical Society before submission of the manuscript to the journal (<https://www.lpi.usra.edu/meteor/>). All meteorite discoveries should in the first instance be submitted for consideration by the Meteorite Nomenclature Committee (at the Committee's website, <https://www.lpi.usra.edu/meteor/naming.php>). You may direct inquiries to the Meteorite Bulletin Editor (metbulleditor@gmail.com): Discovery papers may subsequently be submitted to Meteoritics and Planetary Science.

Existing meteorite names should conform to those in the Meteoritical Bulletin Database (<https://www.lpi.usra.edu/meteor/>). The full (unabbreviated) names of meteorites should be used in titles, headings and at first mention in the text. Abbreviations, including those published in the Antarctic Meteorite Newsletter and the Meteoritical Bulletin Database, may be used in tables and elsewhere. Note that the abbreviated form of numbered meteorites, including Antarctic and many Saharan specimens, should have a blank space between the abbreviated place name and the number (e.g., MacAlpine Hills 88105 is abbreviated MAC 88105), except that some

Antarctic meteorites recovered prior to 1981 may have an "A" instead of the blank space (e.g., Elephant Moraine A79001 is abbreviated EETA79001). A list of standard abbreviations and examples of their proper usage can be found at the Meteoritical Society website page maintained for this purpose (<http://www.lpi.usra.edu/meteor/DenseAreas.php>). Geochimica et Cosmochimica Acta will normally not consider publication of manuscripts purporting to describe the discovery of new meteorites.

When applicable, any available collection numbers for meteorite samples should be provided with the corresponding dataset, and acknowledgements of meteorite loan sources are encouraged. Such additional information may be used to track samples and data sources, to ensure that research on meteorites is reproducible and to prevent confusion in the literature. The article "Best practices for the use of meteorite names in publications" by Heck et al. (2019) (Meteoritics & Planetary Science, 54, 7, 1397-1400) details such requirements for the publication of meteorite research.

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All maps in GCA should display boundaries based solely on physical geography and cannot display national borders or other political boundaries. Some borders are in dispute, and the policy of GCA is to remain neutral in all such disputes. The journal has set a uniform policy, applicable to all authors, that maps cannot display borders. All maps should clearly convey the location or relevant sample locations or study areas. No extraneous maps showing areas or regions irrelevant to the context of the study should be included. The geodetic system that is the basis for a map (e.g., WGS 84) should be conveyed in the figure caption or methods section.

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